```
import java.util.*;

// ===== USER MODEL =====
class User {
    String username;
    String password;
    String role; // USER or ADMIN

    public User(String username, String password, String role) {
        this.username = username;
        this.password = password;
        this.role = role;
    }
}

// ===== AUTH SERVICE =====
class AuthService {
    ArrayList<User> users = new ArrayList<>();

    public AuthService() {
        //   ADMIN ACCOUNT
        users.add(new User("admin", "admin123", "ADMIN"));
    }

    public void register(String username, String password) {
        users.add(new User(username, password, "USER"));
        System.out.println(" Registered successfully!");
    }

    public User login(String username, String password) {
        for (User u : users) {
            if (u.username.equals(username) && u.password.equals(password)) {
                return u;
            }
        }
        return null;
    }

    //  FORGOT PASSWORD
    public void forgotPassword(String username) {
        for (User u : users) {
            if (u.username.equals(username)) {
                System.out.print("Enter new password: ");
                Scanner sc = new Scanner(System.in);
```

```
            u.password = sc.nextLine();
            System.out.println(" Password updated!");
            return;
        }
      }
      System.out.println("Username not found!");
   }
}

// ===== ITEM MODEL =====
class Item {
   String name, description, location, date, type, owner, claimCode;
   boolean claimed = false;

   public Item(String name, String description, String location, String date, String type, String
owner) {
      this.name = name;
      this.description = description;
      this.location = location;
      this.date = date;
      this.type = type;
      this.owner = owner;
      this.claimCode = generateCode();
   }

   private String generateCode() {
      return "CODE" + (int)(Math.random() * 1000);
   }
}

// ===== ITEM SERVICE =====
class ItemService {
   ArrayList<Item> items = new ArrayList<>();

   public void addItem(Item item) {
      items.add(item);
      System.out.println(" Item added!");
   }

   public void viewItems() {
      for (Item i : items) {
         // Displays the date as requested
         System.out.println(i.type + " | " + i.name + " | " + i.location + " | Date: " + i.date + " |
Claimed: " + i.claimed);
```

```
    }
}

public void matchItems() {
    boolean matchFound = false; // Flag to track if any match exists
    for (Item lost : items) {
        if (lost.type.equals("LOST")) {
            for (Item found : items) {
                if (found.type.equals("FOUND") &&
                    lost.name.equalsIgnoreCase(found.name)) {

                    System.out.println(" MATCH FOUND!");
                    System.out.println("Item: " + lost.name);
                    System.out.println("Claim Code: " + found.claimCode);
                    matchFound = true;
                }
            }
        }
    }
    // Added the "No matches found" message here
    if (!matchFound) {
        System.out.println(" No matches found at this time.");
    }
}

public void claimItem(String code) {
    for (Item i : items) {
        if (i.claimCode.equals(code) && !i.claimed) {
            i.claimed = true;
            System.out.println(" Item claimed: " + i.name);
            return;
        }
    }
    System.out.println("Invalid or already claimed.");
}

// ADMIN FEATURES
public void deleteItem(int index) {
    if (index >= 0 && index < items.size()) {
        items.remove(index);
        System.out.println(" Item deleted.");
    } else {
        System.out.println(" Invalid index.");
    }
```

```
    }
}

// ===== MAIN SYSTEM =====
public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        AuthService auth = new AuthService();
        ItemService service = new ItemService();

        while (true) {
            System.out.println("\n=== PHINMA COC findIT: LOST & FOUND ===");
            System.out.println("1. Register");
            System.out.println("2. Login");
            System.out.println("3. Forgot Password");

            int choice = sc.nextInt();
            sc.nextLine();

            if (choice == 1) {
                System.out.print("Username: ");
                String u = sc.nextLine();
                System.out.print("Password: ");
                String p = sc.nextLine();
                auth.register(u, p);
            }

            else if (choice == 2) {
                System.out.print("Username: ");
                String u = sc.nextLine();
                System.out.print("Password: ");
                String p = sc.nextLine();

                User currentUser = auth.login(u, p);

                if (currentUser != null) {
                    System.out.println(" Login successful!");

                    // ============================
                    // ADMIN PANEL
                    // ============================
                    if (currentUser.role.equals("ADMIN")) {
                        while (true) {
```

```
            System.out.println("\n=== ADMIN PANEL ===");
            System.out.println("1. View Items");
            System.out.println("2. Delete Item");
            System.out.println("3. Logout");

            int opt = sc.nextInt();
            sc.nextLine();

            if (opt == 1) service.viewItems();

            else if (opt == 2) {
                service.viewItems();
                System.out.print("Enter index to delete: ");
                int index = sc.nextInt();
                service.deleteItem(index);
            }

            else if (opt == 3) break;
        }
    }

    // ============================
    //  USER PANEL
    // ============================
    else {
        while (true) {
            System.out.println("\n1. Report Lost");
            System.out.println("2. Report Found");
            System.out.println("3. View Items");
            System.out.println("4. Match Items");
            System.out.println("5. Claim Item");
            System.out.println("6. Logout");

            int opt = sc.nextInt();
            sc.nextLine();

            if (opt == 1 || opt == 2) {
                System.out.print("Item Name: ");
                String name = sc.nextLine();
                System.out.print("Description: ");
                String desc = sc.nextLine();
                System.out.print("Location: ");
                String loc = sc.nextLine();
                System.out.print("Date: ");
```

```
                    String date = sc.nextLine();

                    String type = (opt == 1) ? "LOST" : "FOUND";

                    service.addItem(new Item(name, desc, loc, date, type, u));
                }

                else if (opt == 3) service.viewItems();
                else if (opt == 4) service.matchItems();

                else if (opt == 5) {
                    System.out.print("Enter Claim Code: ");
                    String code = sc.nextLine();
                    service.claimItem(code);
                }

                else if (opt == 6) break;
            }
        }

    } else {
        System.out.println(" Invalid login!");
        System.out.println("Redirecting to Register...");

        System.out.print("New Username: ");
        String newU = sc.nextLine();
        System.out.print("New Password: ");
        String newP = sc.nextLine();

        auth.register(newU, newP);
    }
}

else if (choice == 3) {
    System.out.print("Enter username: ");
    String user = sc.nextLine();
    auth.forgotPassword(user);
}
}
}
}
```